

Due: 5:00 pm, May 8, Thursday

Upload a Word file answering the questions and your original Python code to receive credit.

The coding part of the exam focuses on 700 tests. Each test includes an image, penetration depth, applied load, and hardness. The applied load varied to generate micro-sized impressions of different sizes.

1. Plot the histograms of the average pixel values of the images. Show three plots corresponding to the three colors (RGB).
2. Analyze the pixel values and eliminate tests if they are outliers. Discuss your solution.
3. Augment the images seven times to increase the total data. Show augmented images of a single test of your choice to receive credit.
4. Use VGG16 with weights pretrained by ImageNet after replacing its top layer. Also, modify the loss function and activation function if needed. Train the model to determine the hardness from the image. Show the scatter plot of the predicted versus actual values.
5. Design your convolutional neural network and repeat the previous question. Your model does not have to be complex.